HOW THARUN ANSWERED VS. CBSE TOPPER MODEL ANSWER - Tharun

Q.1 to Q.5: Section A: Objective & Assertion-Reasoning [5 Marks]

Tharun's Written Answer	CBSE Topper Model Answer (100% Marks)
Student Script: 1.(b) $(a-b)^2=a^2-2ab+b^2$; 2.(c) Pascal; 3.(c) Drip Irrigation; 4.(c) Permanent Settlement; 5.(a) Both A and R are true and R explains A. Slip/Gap: None. Flawless recall.	Topper Model Answer: Option letter followed by full text correctly stated. Keywords to Underline: Identity $(a-b)^2$, Pascal, Drip Irrigation, Permanent Settlement

Q.6: Factorise by grouping: 15xy - 6x + 5y - 2 [2 Marks]

Tharun's Written Answer	CBSE Topper Model Answer (100% Marks)
Student Script: = 3x(5y - 2) + 1(5y - 2) = (5y - 2)(3x + 1) Slip/Gap: Good layout. Always write '=' at start of each line.	Topper Model Answer: 15xy - 6x + 5y - 2 = 3x(5y - 2) + 1(5y - 2) = (5y - 2)(3x + 1). Boxed final answer. Keywords to Underline: Common factor (5y - 2)

Q.9: Calculate maximum pressure exerted by wooden block 20x10x5 cm of mass 4 kg [3 Marks]

Tharun's Written Answer	CBSE Topper Model Answer (100% Marks)
Student Script: F = m*g = 4*10 = 40 N. Min Area = 10cm * 5cm = 0.10 * 0.05 = 0.005 m^2. P = F/A = 40 / 0.005 = 8000. Slip/Gap: Omitted S.I. unit next to 8000. CBSE spot-evaluators deduct 0.5 mark immediately.	Topper Model Answer: Maximum Pressure (P) = Force / Minimum Area = 40 N / 0.005 m^2 = 8000 Pa (or N/m^2). Boxed: [P_{max} = 8,000 Pa]. Keywords to Underline: P = F / A, Minimum Surface Area, 8,000 Pa

Q.10: Perimeter of pool is 154 m. Length is 2m more than 2b. Find l and b [3 Marks]

Tharun's Written Answer	CBSE Topper Model Answer (100% Marks)
Student Script: Let b = x m. l = (2x + 2) m. P = 2(l + b) => 154 = 2(2x + 2 + x) => 154 = 2(3x + 2) => 77 = 3x + 2 => 3x = 75 => x = 25 m. Breadth = 25 m, Length = 2(25)+2 = 52 m. Slip/Gap: None. Excellent work.	Topper Model Answer: Breadth = 25 m, Length = 52 m. Verification: 2(52 + 25) = 2(77) = 154 m. Keywords to Underline: Let breadth = x m, Perimeter = 2(l + b), 25 m, 52 m

Q.12: Cylindrical storage tank radius 7m, height 3m. Sheet required & cost [5 Marks]

Tharun's Written Answer	CBSE Topper Model Answer (100% Marks)
Student Script: Total Surface Area = 2*pi*r*(r + h) = 2 * (22/7) * 7 * (7 + 3) = 44 * 10 = 440 m^2. Cost = 440 * 45 = ☐ 19,800. Slip/Gap: Calculation was correct but missed the formal concluding statement required in 5M CBSE questions.	Topper Model Answer: 1. TSA = $2\pi r(r + h)$ = 440 m^2. 2. Total Cost = Area x Rate = 440 x 45 = ☐ 19,800. Statement: Hence, 440 m^2 of sheet is needed at total cost of ☐ 19,800. Keywords to Underline: Total Surface Area = $2\pi r(r+h)$, 440 m^2, ☐ 19,800